Math 63: Real Analysis

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1 Differentiation

- **Definition:** A real valued function f on an open subset $U \subset \mathbb{R}$ is differentiable at $x_0 \in U$ if
$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$
exists. That is for each $\epsilon > 0$ there exists $\delta > 0$ such that
$$\left| \frac{f(x) - f(x_0)}{x - x_0} - f'(x_0) \right| \leq \epsilon \implies |f(x) - f(x_0) - f'(x_0)(x - x_0)| \leq \epsilon |x - x_0|$$
whenever $|x - x_0| < \delta$.

- f differentiable $\implies$ f continuous.

- Let U be an open susbset of $\mathbb{R}$. If $f : U \to \mathbb{R}$ attains a maximum/minimum at $x_0 \in U$ then $f'(x_0) = 0$.

- **Rolle's Theorem:** Let $a, b \in \mathbb{R}, a < b$ and let $f : [a, b] \to \mathbb{R}$ continuous, and differentiable on (a, b), and such that $f(a) = f(b) = 0$. Then there exists $c \in (a, b)$ such that $f'(c) = 0$.

  Proof: Since $[a, b]$ compact, f attains a max and a min. If max and min @ end points, then $f'(x) = 0$ for all x. Else by the statement above, there exists an x_0 such that $f'(x_0) = 0$.

- **MVT:** Let $a, b \in \mathbb{R}, a < b$ and let $f : [a, b] \to \mathbb{R}$ continuous, and differentiable on (a, b), and such that $f(a) = f(b) = 0$. Then there exists $c \in (a, b)$ such that
$$f(b) - f(b) = (b - a)f'(c)$$

- If f has positive/negative derivative at each point then f' is strictly inc/dec.

- If $f : U \to \mathbb{R}$ is $(n+1)$ times differentiable, then for any $a, b \in U$ we define $R_n(b, a)$ by,
$$f(b) = f(a) + \frac{f'(a)(b-a)}{1!} + \frac{f''(a)(b-a)^2}{2!} + \cdots + \frac{f^{(n)}(a)(b-a)^n}{n!} + R_n(b, a)$$

and

$$\frac{d}{dx}R_n(b,x) = -\frac{f^{(n+1)}(x)(b-x)^n}{n!}$$

- **(not) Taylor's Thm :** If $f : U \to \mathbb{R}$ is $(n+1)$ times differentiable, then for any $a, b \in U$ we define $R_n(b,a)$ by,

$$f(b) = f(a) + \frac{f'(a)(b-a)}{1!} + \frac{f''(a)(b-a)^2}{2!} + \cdots + \frac{f^{(n)}(a)(b-a)^n}{n!} + \frac{f^{(n+1)}(c)(b-a)^{n+1}}{(n+1)!}$$

  where $c \in [a,b]$.

2 Integration

- **Definition:** Partition of a closed interval $[a,b]$ refers to a finite sequence of numbers such that $a = x_0, x_1, \ldots, x_N = b$. The width of the partition is

$$\max\{x_i - x_{i-1}, i = 1, 2, \ldots, N\}$$

  And the Riemann sum for $f : [a,b] \to \mathbb{R}$ corresponding to a given partition is

$$S = \sum_{i=1}^{N} f(x_i')(x_i - x_{i-1})$$

  where $x_i' \in (x_{i-1}, x_i)$. And f is Riemann integrable if there exists $A \in \mathbb{R}$ such that for any $\epsilon > 0$ there exists a $\delta > 0$ such that $|S - A| < \epsilon$ where S is a Riemann sum corresponding to a partition of $[a,b]$ of width less than δ. Then $A = \int_a^b f(x)dx$.

- **"Cauchy-like" criterion:** $f : [a,b] \to \mathbb{R}$ is Riemann integrable iff $\forall \epsilon > 0$ there exists $\delta > 0$ such that $|S_1 - S_2| < \epsilon$ whenever S_1, S_2 are Riemann sums for f of width less than δ.

- Step-functions are integrable, and $f : [a,b] \to \mathbb{R}$ is integrable iff for each $\epsilon > 0$ there exist step functions f_1, f_2 on $[a,b]$ such that $f_1(x) \le f(x) \le f_2(x)$ for each $x \in [a,b]$, and $\int_a^b (f_2(x) - f_1(x))dx < \epsilon$.

- If $f : [a,b] \to \mathbb{R}$ is integrable, then it is bounded.

- **FTC:** Let $f : U \to \mathbb{R}$ continous, $a \in U$. Define $F : U \to \mathbb{R}$ such that $F(x) = \int_a^x f(t)dt$ for all $x \in U$. Then F is differentiable and $F' = f$.

3 Interchange of Limit Operations

- **Thm:** If $f_1, f_2, f_3, \ldots$ is a uniformly convergent sequence of continous real-valued funcs on $[a, b]$ then
$$\int_a^b \left(\lim_{n\to\infty} f_n(x)\right) dx = \lim_{n\to\infty} \int_a^b f_n(x) dx$$

- **Thm:** Let $f_1, f_2, f_3, \ldots$ be a sequence of real-valued functions on an open interval $U \in \mathbb{R}$, each having a continuous derivative. Suppose that the sequence $f_1', f_2', f_3', \ldots$ converges untformly on U and that for some $a \in U$ the sequence $f_1(a), f_2(a), f_3(a), \ldots$ converges. Then $\lim\limits_{n\to\infty}$ exists, is differentiable, and
$$\left(\lim_{n\to\infty} f_n\right)' = \lim_{n\to\infty} f_n'$$

- $\sum_{n=1}^{\infty} a_n$ converges iff for any $\epsilon > 0$ there is a positive integer N such that if $n \geq m \geq N$ then $|a_{m+1} + a_{m_2} + \ldots + a_n| < \epsilon$.

- If $\sum_{i=1}^{\infty} a_n$ converges then $\lim\limits_{n\to\infty} a_n = 0$.

- If $a_1, a_2, a_3, \ldots$ are nonnegative real numbers, then either the series $\sum_{n=1}^{\infty} a_n$ converges or it has arbitrarily large partial sums.

- **Definition:** If $a_1, a_2, a_3, \ldots$ are real numbers, the series $\sum_{n=1}^{\infty} a_n$ is said to be absolutely convergent if the series $\sum_{n=1}^{\infty} |a_n|$ is convergent.